

CLARKSVILLE OUTLAW, TN, USA

WMO: 723280

Lat: 36.624N

Lon: 87.419W

Elev: 171

StdP: 99.29

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.1	-17.2	0.8	-8.2	-14.5	1.1	-6.8	11.0	5.4	9.9	5.7	3.0	340	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.1	23.8	32.9	24.0	32.1	23.7	25.7	31.6	25.2	30.9	24.7	29.9	3.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.3	29.0	23.5	18.7	28.3	22.9	18.0	27.7	80.2	31.7	77.7	30.9	75.3	30.0	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.0	7.9	7.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-14.2	36.4	3.4	2.7	-16.7	38.3	-18.7	39.9	-20.6	41.4	-23.1	43.3
			-14.9	26.9	3.2	0.9	-17.2	27.5	-19.0	28.1	-20.8	28.6	-23.1	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	3.1	4.6	9.7	15.0	19.9	24.3	25.5	25.3	21.9	15.5	8.9	4.9
	DBStd	9.44	6.66	6.51	6.03	4.96	4.00	2.61	2.45	2.67	3.52	4.82	5.17	5.74
	HDD10.0	767	231	174	83	14	0	0	0	0	0	8	79	179
	HDD18.3	2112	473	385	274	121	33	1	0	1	10	114	283	417
	CDD10.0	2569	16	23	72	163	307	430	481	473	357	179	47	20
	CDD18.3	872	0	1	5	20	82	181	223	215	117	27	2	1
	CDH23.3	8417	0	6	42	199	705	1791	2212	2106	1076	270	11	0
	CDH26.7	3327	0	1	4	26	193	724	948	943	418	68	1	0
Wind	WSAvg	2.8	3.6	3.6	3.6	3.4	2.6	2.3	2.1	1.9	2.0	2.6	2.9	3.2
Precipitation	PrecAvg	1237	93	102	128	107	118	105	98	88	85	79	111	123
	PrecMax	1602	247	274	404	229	295	282	229	229	294	196	226	318
	PrecMin	817	21	31	46	21	20	12	29	16	11	1	32	25
	PrecStd	187	50	54	69	49	57	57	45	51	55	46	48	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	23.2	27.1	29.0	31.4	34.9	35.7	37.5	34.6	31.1	24.7	21.9
		MCWB	15.7	16.1	17.8	18.9	22.0	22.3	24.7	23.4	22.6	21.0	17.0	17.5
	2%	DB	17.8	20.3	24.2	27.3	30.1	32.8	33.6	34.8	32.7	28.5	22.2	19.0
		MCWB	14.1	15.6	16.6	17.9	21.4	23.8	24.8	23.6	22.4	19.8	15.8	15.4
	5%	DB	16.2	17.9	22.0	25.8	28.9	31.9	32.6	32.9	31.1	26.4	19.8	16.5
		MCWB	13.1	13.3	15.4	17.3	20.7	23.7	24.5	23.9	22.4	18.5	14.5	13.6
	10%	DB	13.9	15.2	19.6	24.0	27.5	30.8	31.5	31.6	29.1	24.1	17.6	14.1
		MCWB	10.6	11.2	13.8	16.5	20.0	23.1	24.0	23.4	21.4	17.8	13.6	10.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	17.7	19.0	21.2	23.7	26.0	26.4	26.7	24.9	22.9	19.2	18.3
		MCDB	19.1	20.4	24.4	26.6	28.9	31.8	32.2	33.1	30.4	28.5	22.4	20.9
	2%	WB	15.2	16.5	17.6	19.7	22.7	25.3	25.8	25.6	24.0	21.7	17.5	16.4
		MCDB	17.3	20.7	22.9	24.9	28.1	31.1	31.6	31.5	29.7	26.5	20.0	18.7
	5%	WB	13.7	14.4	16.4	18.6	21.9	24.6	25.3	25.0	23.2	20.4	15.7	14.3
		MCDB	16.0	17.1	20.4	23.4	27.2	30.1	31.1	30.4	28.7	24.3	18.8	16.4
	10%	WB	11.1	11.8	14.9	17.5	21.1	23.9	24.8	24.3	22.4	19.0	13.9	11.4
		MCDB	13.5	14.6	19.2	22.2	25.8	29.1	30.0	29.2	27.5	22.8	16.9	13.6
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.1	11.5	12.2	11.1	11.2	10.6	11.5	12.1	12.3	11.5	9.4
		MCDBR	11.9	12.8	14.0	14.0	12.6	12.3	11.8	13.4	13.9	14.7	14.1	11.5
		MCWBR	9.3	9.0	8.1	7.0	5.5	4.8	4.5	4.8	5.2	6.9	8.6	8.8
	5% WB	MCDBR	11.0	11.1	11.5	11.0	10.6	10.5	10.4	10.7	11.2	11.6	10.8	10.3
		MCWBR	9.6	8.8	7.3	6.4	5.0	4.7	4.5	4.6	4.8	6.3	8.0	8.9
Clear-Sky Solar Irradiance	taub	0.316	0.323	0.372	0.400	0.441	0.460	0.497	0.499	0.423	0.366	0.330	0.309	
	taud	2.479	2.439	2.346	2.267	2.203	2.176	2.093	2.076	2.280	2.424	2.479	2.506	
	Ebn at Noon	878	913	893	882	847	828	794	783	829	852	853	862	
	Edh at Noon	83	97	117	133	144	148	160	159	123	97	82	76	
All-Sky Solar Radiation	RadAvg	2.17	2.75	3.84	5.01	5.69	6.32	6.05	5.56	4.68	3.58	2.53	1.84	
	RadStd	0.19	0.35	0.31	0.33	0.49	0.43	0.35	0.26	0.33	0.40	0.26	0.18	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	+76

Nomenclature: See separate page